Claudio **Tomasi**

Viale Matteotti 73, 27100, Pavia

(+39) 3401268139 | claudio.tomasi93@gmail.com | claudiotomasi | claudiotomasi

Research Interests

Transport modeling and accessibility analysis, Optimization algorithms, Combinatorial optimization, Data structures, Algorithms on graphs, High-performance programming, Data-driven models, Geo-spatial data analyses, Data-driven policy support and decision analysis.

Research Positions

Joint Research Center of the European Commission - Unit C6 *Seville, Spain (Remote)*

EXTERNAL EXPERT *Apr. 2025 - Now*

- Enhancements of in-house public transport schedule-based routing tools

Department of Economical Sciences, Università degli Studi di Bergamo *Bergamo, Italy*

POSTDOCTORATE FELLOW *Aug. 2024 - Now*

- Participation in HEXAGON project: Exact Algorithms for Power Grids

Joint Research Center of the European Commission - Unit C6 *Seville, Spain*

TRANSPORT POLICY RESEARCHER *Sep. 2023 - May. 2024*

- Modeling large-scale graph to simulate road networks and assess accessibility to electric recharging points.
- Coverage analysis of electric vehicle charger infrastructure in EU road network, using TENtec data.
- Study and analysis of performance of public and private transport in EU member states, to achieve accessibility to services.

Department of Mathematics "F. Casorati", Università degli Studi di Pavia *Pavia, Italy*

POSTDOCTORATE FELLOW *Mar. 2022 - Ago. 2023*

- Participation in European Research Project: Development of a multimodal schedule-based route planner

Education

Università della Svizzera Italiana (USI) & Università degli Studi di Pavia *Lugano, Switzerland & Pavia, italy*

JOINT PHD IN COMPUTATIONAL MATHEMATICS AND DECISION SCIENCES *Oct. 2018 - Feb. 2022*

- One year spent at USI for reasearch activity
- Dissertation title: "Construction of Grid Operators for Multilevel Solvers: a Neural Network Approach"

Università degli Studi di Milano Bicocca & Università della Svizzera Italiana (USI) *Milano, Italy & Lugano, Switzerland*

DOUBLE MASTER'S DEGREE *Nov. 2016 - Oct. 2018*

- First Year at Università degli Studi di Milano - Bicocca Second Year at Università della Svizzera Italiana (USI) - Lugano
- Final dissertation title: "Uncertainty quantification methods for random fracture networks"
  Vote: 110L/110

Università degli Studi di Milano Bicocca *Milano, Italy*

BACHELOR'S DEGREE IN COMPUTER SCIENCE *Sep. 2013 - Jul. 2016*

- Final dissertation title:"Closure Operators for Cyclic and Concurrent Processes". Vote: 109/110

Publications

Forthcoming Works

- D. Duma, et al., *VelociRAPTOR: A new algorithm for multimodal all-pairs time-dependent shortest paths at European scale.*.
- G. Giordano, et al., *Investigating access to services and opportunities in urban and rural environments: the case studies of Estonia and the Netherlands*.
- S. Coniglio, et al., *Bilevel optimization methods for transformer-rating optimization.*.

Journal Articles

- C. Tomasi, R. Krause, *Construction of Grid Operators for Multilevel Solvers: a Neural Network Approach*. In: "Domain Decomposition Methods in Science and Engineering XXVI", Springer LNCSE Series 145, 2023

Technical Reports

- D. Duma, et al. *VelociRAPTOR Reference Manual*, 2023

Teaching Activities

Adjunct professor *Pavia, italy*

DEPARTMENT OF ENGINEERING - UNIVERSITÀ DEGLI STUDI DI PAVIA *Oct. 2022 - Now*

- Course of Applied Informatics (*3 years*).

Teaching Assistant
Lugano, Switzerland

Department of Informatics - Università della Svizzera Italiana — *2019 - 2020*

- Course of Databases of M.Sc in Management and Informatics (*1 year*).
- Course of Algorithm & Complexity of M.Sc in Informatics (*1 year*).

Workshops & Conferences

2024	**Attended Conference**, The HEXAGON Workshop on power grids	*Bergamo, Italy*
2023	**Co-Author of Contributed Talk**, ODS 2023 - Conference on Optimization and Decision Sciences	*Ischia, Italy*
2023	**Co-Author of Contributed Talk**, TDM2023: 11th International Travel Demand Management Symposium	*Kumasi, Ghana*
2023	**Co-Author of Contributed Talk**, ORAHS (Operational Research Applied to Health Services) 2023	*Graz, Austria*
2023	**Contributed talk**, 7th AIRO YOUNG Workshop	*Milan, Italy*
2022	**Co-Author of Contributed Talk**, ODS 2022 - Conference on Optimization and Decision Sciences	*Florence, Italy*
2022	**Invited talk**, CompMat Spring Workshop 2022	*Pavia, Italy*
2021	**Invited talk**, Se mi narri di Matematica	*Pavia, Italy*
2020	**Invited talk**, The 26th International Domain Decomposition Conference (DD26)	*Hong Kong, China*
2019	**Invited talk**, PhD Spring Workshop: Computational Mathematics, Statistics and Machine Learning	*Pavia, Italy*

Projects

Development of a multimodal schedule-based route planner
Pavia, Italy

Participation in European Research Project — *Dec. 2021 - Now*

- C++ software project to efficiently solve multimodal all-pairs shortest path problems on large instances as EU countries
- Modeling and analysis of multimodal networks: direct routing on the timetable for public transportation systems, and graph modeling of the road networks.
- High-performing parsing, managing and analyze large public transport datasets, to perform operations and find solutions on the resulting underlying graphs
- Density maps of EU state members, showing the number of reached public stops using different modes of transport.
- Accessibility maps of EU state members, showing the number of reached services, e.g., schools and hospitals
- Collaboration with Joint Research Center (JRC) based in Seville

A Neural Network approach for the generation of Transfer Operators in Multilevel Solvers
Pavia, Italy

PhD Thesis Project — *Dec. 2021*

- Python project using Neural Network models to predict operators of Multigrid methods in solving discretized PDEs
- Definition and generation from scratch of the needed dataset, in order to train the Neural Network
- Study of data distribution of the dataset to improve the learning ability of the model
- Visualization of specific physical domains to graphically study and show the solution of a given PDE.
- Source code available at the public Bitbucket repository "Learn Multigrid"

Multilevel Monte Carlo Methods for Random Fracture Networks
Lugano, Switzerland

Master's Thesis (Participation in FASTER Project) — *Sep. 2018*

- Simulations of fluid propagation in a porous underground, in order to understand if a given domain can be used to produce geothermal energy and if there is an earthquake possibility.
- Analysis of real scenarios, to then generate plausible data for creating new scenarios
- Visualization of physical domains to understand the distribution of the rock fractures created by the fluid during the simulations.
- To make the simulations faster, the MLMC method is used, also providing an estimate of the statistical error

Skills

Programming	Python, C++, R, MATLAB, Java, HTML, CSS, PHP, C, Ruby. Multithreading
Data Analyses & Modeling	Pandas, Scikit-learn, Tensorflow, Clustering, Scenario Simulation
Transport & Geospatial Tools	OpenTripPlanner, QGIS, Geopandas, Folium, Leaflet, Matplotlib. Spatial Analysis
Optimization & Algorithms	Graph algorithms, Combinatorial optimization, Multimodal routing, Multigrid methods
Databases	ER and Relational Modelling, MySQL
Software	Git, LaTeX, Mendeley, Zotero, Microsoft Office
Languages	Italian (native), English (fluent)